

Machine Learning is a subset of Artificial Intelligence that focuses on building systems that learn from data. Linear Algebra is the mathematical foundation for Machine Learning, including vectors and matrices. Neural Networks are inspired by the human brain and consist of interconnected layers of nodes. Backpropagation is used to train Neural Networks using the chain rule from Calculus. Gradient Descent is an optimization algorithm used to minimize the Loss Function by adjusting weights. Stochastic Gradient Descent is a variant of Gradient Descent that updates weights using a single sample. Convolutional Neural Networks are deep neural networks designed for grid data like images. Recurrent Neural Networks process sequential data like text. Attention Mechanism dynamically focuses on specific parts of the input sequence. Transformers rely solely on Attention Mechanism, replacing recurrence and convolutions. Graph Neural Networks perform inference on graph-structured data. For example, you can build a Secure Online Voting System using Blockchain and Zero-Knowledge Proofs.